Exploratory Data Analysis on Diwali Sales Dataset

Objective

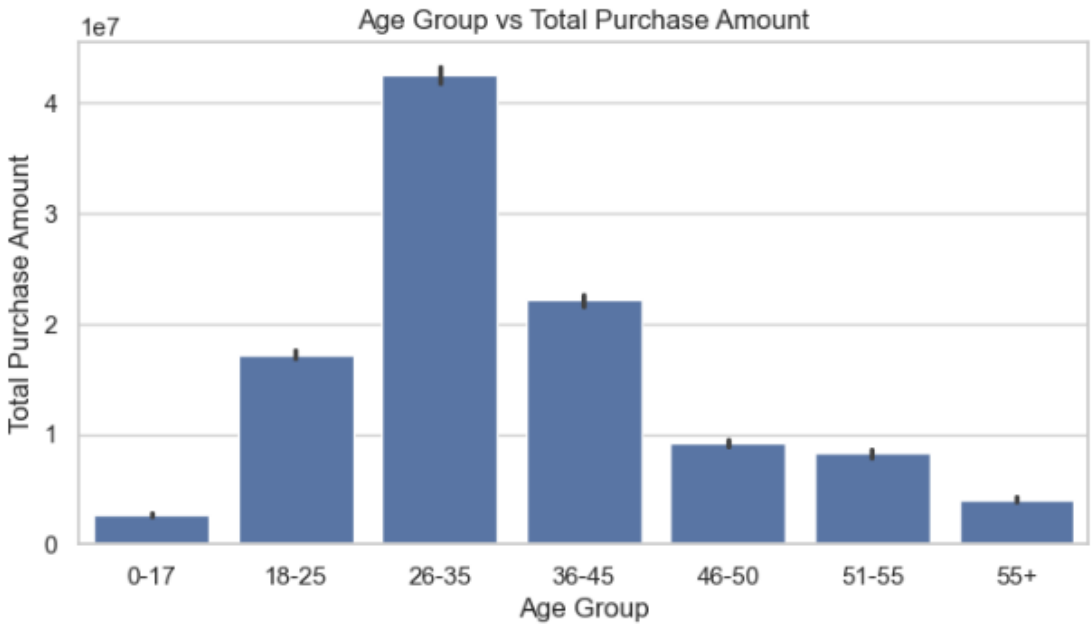
The objective of this project is to analyze the Diwali Sales dataset using Exploratory Data Analysis (EDA) techniques in order to understand customer purchasing behavior, identify popular product categories, and study sales trends across different regions.

Tools and Libraries Used

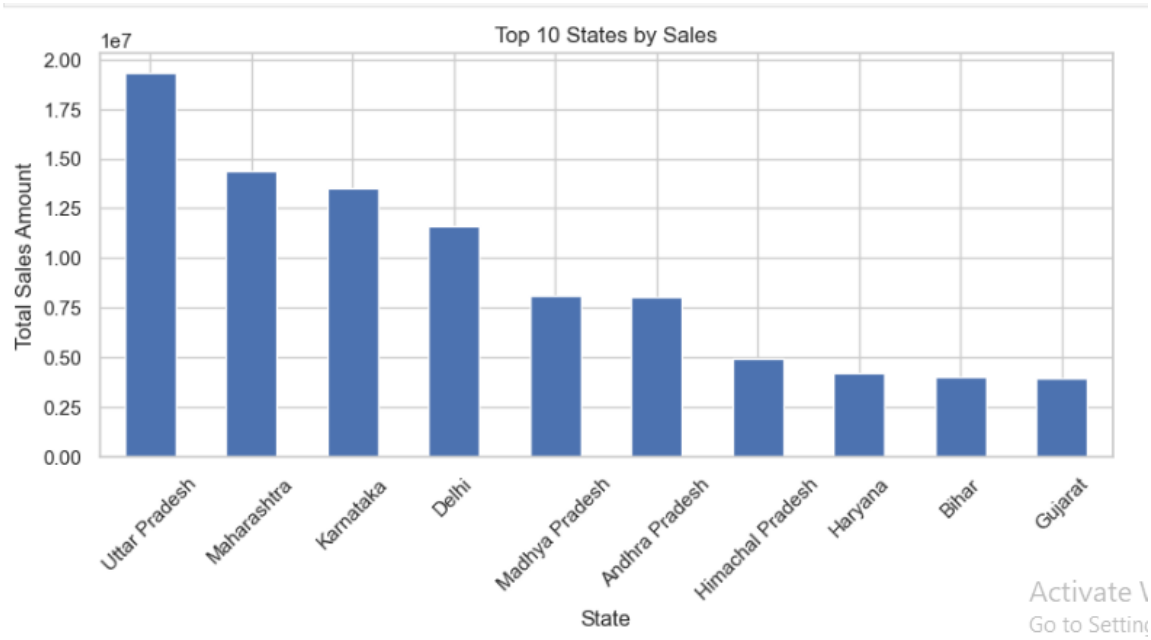
- Python – for data analysis
- Pandas – for data manipulation and cleaning
- NumPy – for numerical operations
- Matplotlib – for data visualization
- Seaborn – for advanced statistical plots

Summary of Insights

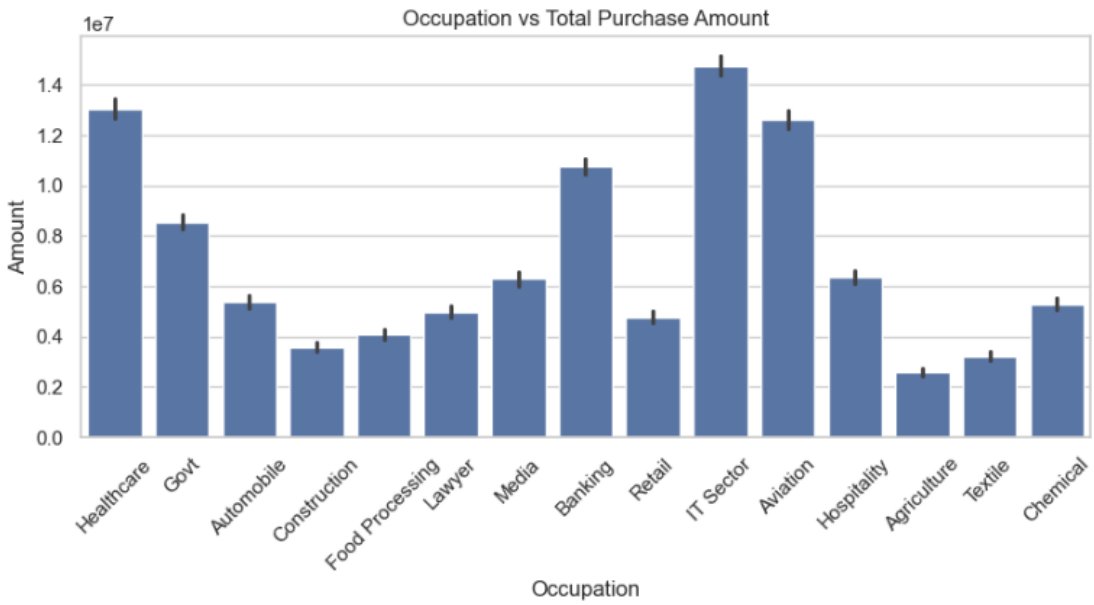
Female customers generated higher total sales compared to male customers during the Diwali season.



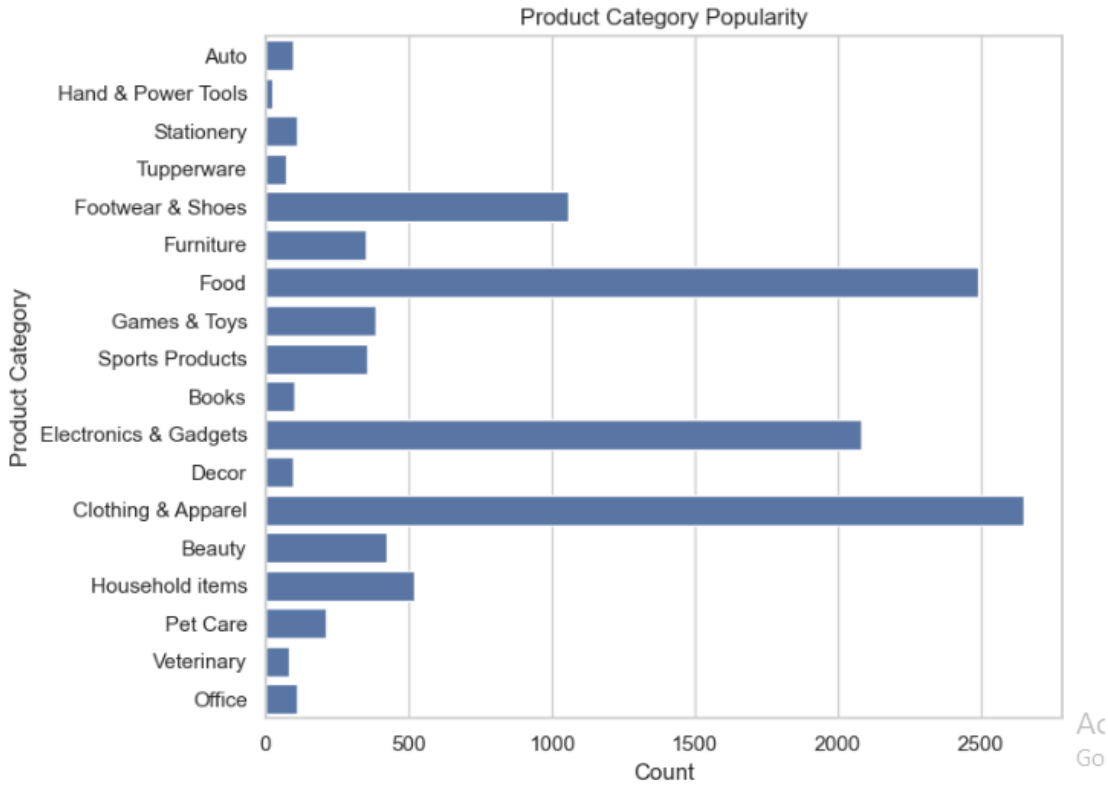
The age group of 26–35 years contributed the highest purchase amount, showing strong buying power.



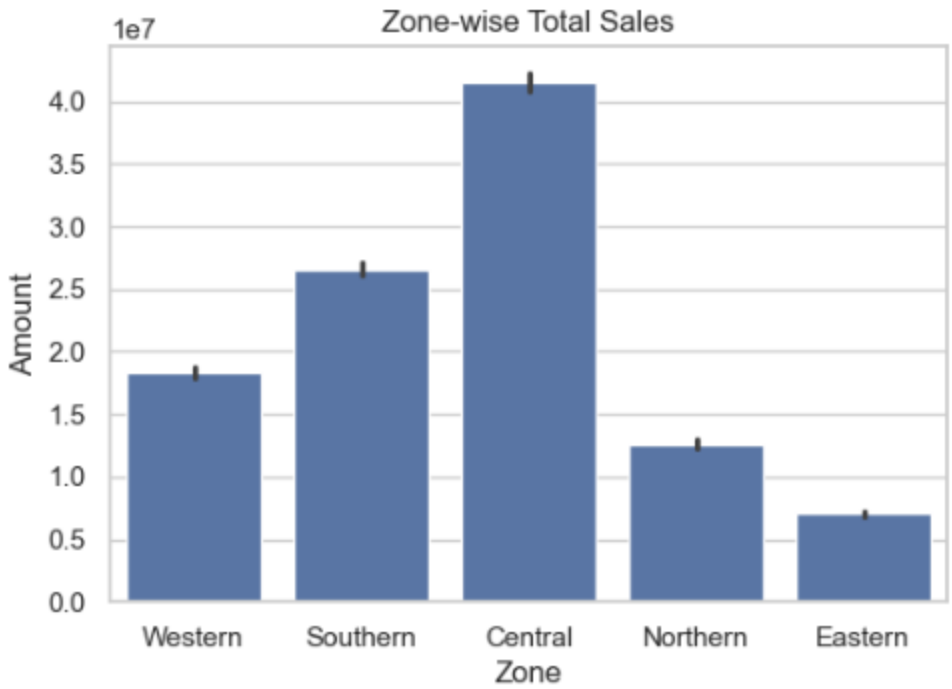
Uttar Pradesh, Maharashtra, and Karnataka emerged as the top states in terms of sales.



Professionals working in IT, Healthcare, and Aviation sectors showed higher spending patterns.



Product categories such as Clothing & Apparel, Food, and Electronics were the most popular.



Central and Southern zones recorded the highest overall sales among all regions.

Challenges Faced and Solutions

During the analysis, the dataset had encoding issues, missing values, and overlapping labels in some visualizations. These problems were solved by using the correct file encoding, removing incomplete records, and adjusting chart sizes and label orientations for better readability.

Conclusion

This exploratory data analysis provided clear insights into customer behavior and sales trends during the Diwali festival. The findings can help businesses make better decisions related to marketing, inventory planning, and regional sales strategies.

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